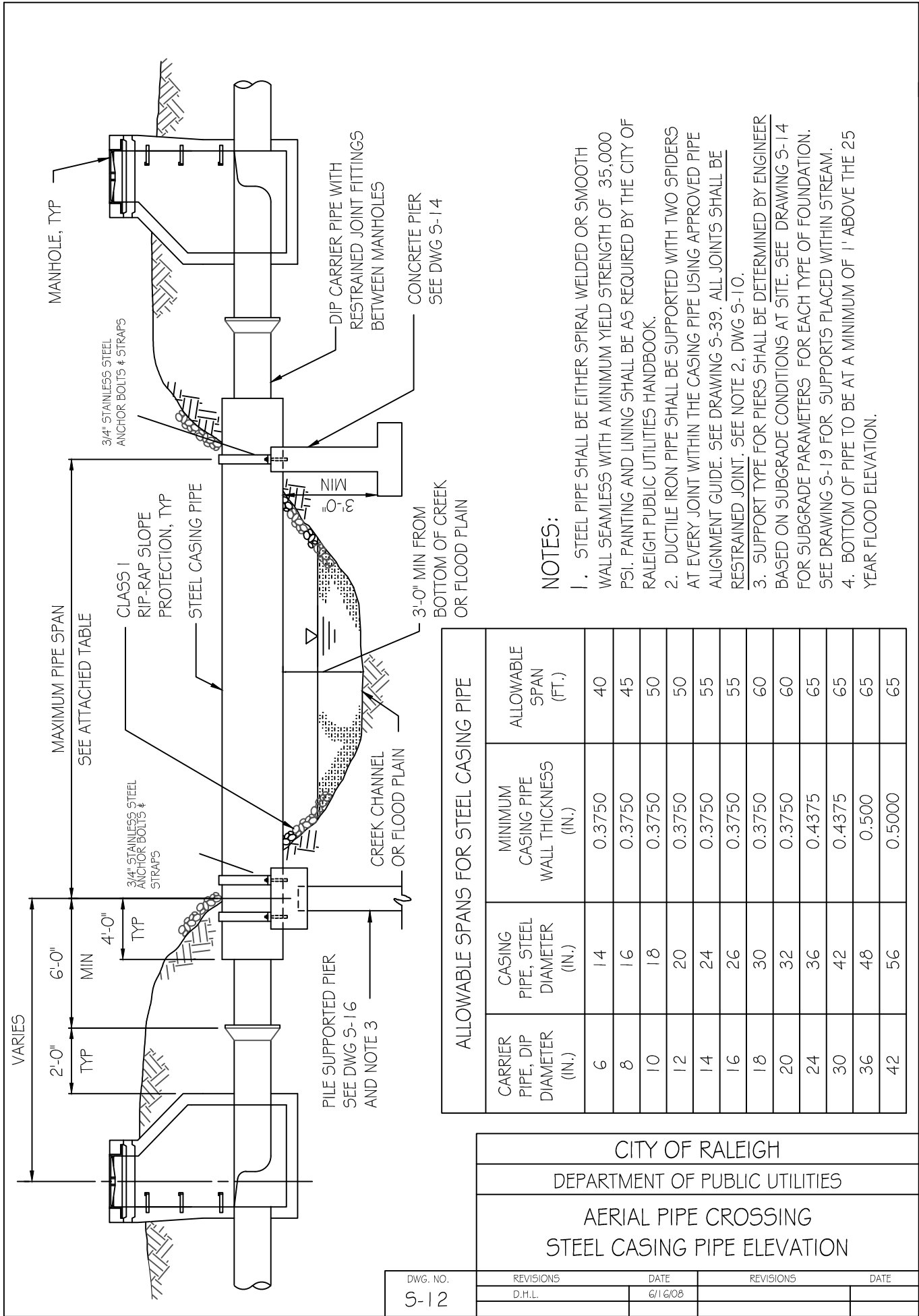




NOTES:

- 1. STEEL PIPE SHALL BE EITHER SPIRAL WELDED OR SMOOTH WALL SEAMLESS WITH A MINIMUM YIELD STRENGTH OF 35,000 PSI. PAINTING AND LINING SHALL BE AS REQUIRED BY THE CITY OF RALEIGH PUBLIC UTILITIES HANDBOOK.
- 2. DUCTILE IRON PIPE SHALL BE SUPPORTED WITH TWO SPIDERS AT EVERY JOINT WITHIN THE CASING PIPE USING APPROVED PIPE ALIGNMENT GUIDE. SEE DRAWING S-39. ALL JOINTS SHALL BE RESTRAINED JOINT. SEE NOTE 2, DWG S-10.
- 3. SUPPORT TYPE FOR PIERS SHALL BE DETERMINED BY ENGINEER BASED ON SUBGRADE CONDITIONS AT SITE. SEE DRAWING S-14 FOR SUBGRADE PARAMETERS FOR EACH TYPE OF FOUNDATION. SEE DRAWING S-19 FOR SUPPORTS PLACED WITHIN STREAM.
- 4. BOTTOM OF PIPE TO BE AT A MINIMUM OF 1' ABOVE THE 25 YEAR FLOOD ELEVATION.

| ALLOWABLE SPANS FOR STEEL CASING PIPE |                                   |                                          |                      |
|---------------------------------------|-----------------------------------|------------------------------------------|----------------------|
| CARRIER PIPE, DIP DIAMETER (IN.)      | CASING PIPE, STEEL DIAMETER (IN.) | MINIMUM CASING PIPE WALL THICKNESS (IN.) | ALLOWABLE SPAN (FT.) |
| 6                                     | 14                                | 0.3750                                   | 40                   |
| 8                                     | 16                                | 0.3750                                   | 45                   |
| 10                                    | 18                                | 0.3750                                   | 50                   |
| 12                                    | 20                                | 0.3750                                   | 50                   |
| 14                                    | 24                                | 0.3750                                   | 55                   |
| 16                                    | 26                                | 0.3750                                   | 55                   |
| 18                                    | 30                                | 0.3750                                   | 60                   |
| 20                                    | 32                                | 0.3750                                   | 60                   |
| 24                                    | 36                                | 0.4375                                   | 65                   |
| 30                                    | 42                                | 0.4375                                   | 65                   |
| 36                                    | 48                                | 0.500                                    | 65                   |
| 42                                    | 56                                | 0.5000                                   | 65                   |

|                                                     |           |         |           |
|-----------------------------------------------------|-----------|---------|-----------|
| CITY OF RALEIGH                                     |           |         |           |
| DEPARTMENT OF PUBLIC UTILITIES                      |           |         |           |
| AERIAL PIPE CROSSING<br>STEEL CASING PIPE ELEVATION |           |         |           |
| DWG. NO.<br>S-12                                    | REVISIONS | DATE    | REVISIONS |
|                                                     | D.H.L.    | 6/16/08 |           |